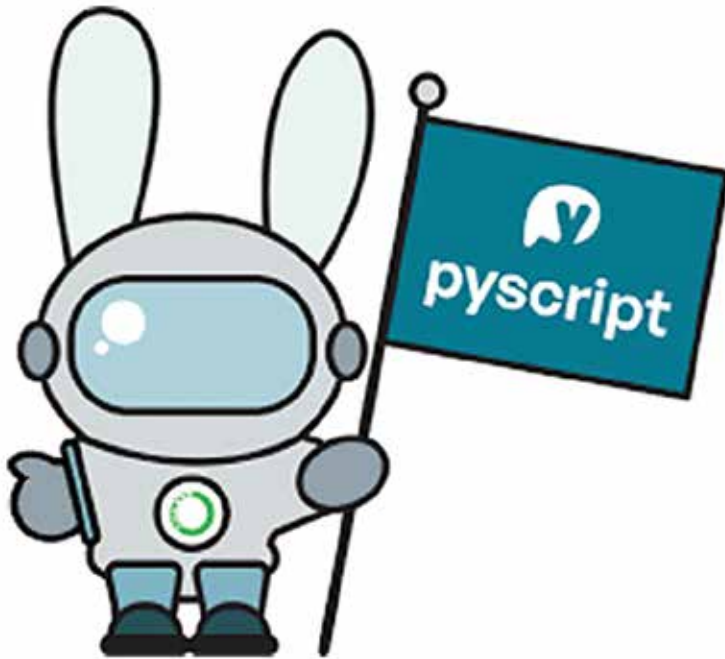


An Introduction to PyScript

Image Source : <https://pyscript.com/>



An open source framework introduced by Anaconda, PyScript is built to run Python from a browser. Though it still has performance issues and a few bugs, it is important to get familiar with it as it promises to be popular in the future.

We are aware that JavaScript client-side coding can be used for server-side coding. Let's now discuss the reverse — using a server-side language from the browser with the help of Python.

But why would one want to execute server-side programs in the browser? The answer is: for enhanced performance, features and convenience. In certain use cases running a code in C is much faster than using JavaScript -- for example, for generating quick reports, dealing with heavy prime numbers, image processing or computer vision on-the-fly using the web page, and so on.

PyScript is a framework built to run Python from a browser; in simple terms, it can see Python code as part of HTML page content. PyScript allows the use of CPython based packages like Pandas and NumPy. It provides bi-directional communication between Python and JavaScript objects and namespaces.

Note that this is not something new in the history of Python. There are already modules that support browser side coding such as Brython, Skulpt, PyJS, and a few more. Everything in this world has its own merits and demerits. Brython does support packages if they are written entirely

in Python. However, programs that use extensions written in C are not supported. For instance, NumPy, Matplotlib, and Pandas cannot work with Brython. Skulpt, on the other hand, does not support a few Python features such as import of additional Python modules that are not available by default.

PyScript has been launched by Anaconda and was officially announced at PyCon US 2022.

How PyScript works

As rightly stated by Anaconda, its Pyodide, Emscripten and WebAssembly (WASM) that create the magic of PyScript. So let's look at each of these and see how they help in building PyScript.

WASM: As we all know, computers understand only binaries, and the same goes for JavaScript as well. A JavaScript engine is needed to read the code and convert it into binary instructions.

WebAssembly or WASM is a language for the web, which is used to code in low level binary format. It can run in modern web browsers.

When WebAssembly was originally developed in 2017 by Mozilla, web browsers only supported JavaScript. It quickly